

Magnetometer-Calibrated Hybrid Transformer for Robust Inertial Tracking in Robotics

Xinzhe Zheng, Sijie Ji, Yipeng Pan, Kaiwen Zhang, Jia Pan, Chenshu Wu*

Abstract—Inertial tracking is vital for autonomous robots and has gained popularity with the ubiquity of low-cost Inertial Measurement Units (IMUs) and deep learning-powered tracking algorithms. Existing works, however, have not fully utilized IMU measurements, particularly magnetometers, nor maximized the potential of deep learning to achieve the desired accuracy. To bridge the gap, we introduce NEURIT, which employs a Time-Frequency Block-recurrent Transformer (TF-BRT) at its core, combining RNN and Transformer to learn both time-frequency representative features. To fully utilize IMU information, we strategically employ differentiation of body-frame magnetometers for orientation calibration in a sensor fusion manner. Experiments conducted in diverse environments show that NEURIT maintains a mere 1-meter tracking error over a 300-meter distance, surpassing state-of-the-art baselines by 48.21% on unseen data. NEURIT also performs comparably to the visual-inertial approach (Tango Phone) in vision-favored conditions and surpasses it in plain environments. We share the code and data to promote further research: <https://github.com/aiot-lab/NeurIT>.

I. INTRODUCTION

Indoor tracking is vital for robots, as it allows them to provide location-based services under any circumstances, or as an auxiliary approach when Vision-based methods [1] degrade in plain environments and bad lighting conditions.

However, achieving high accuracy in inertial tracking is challenging due to sensor noise and drifts. Traditional filtering control algorithms [2], [3] offer noise reduction, yet their performance degrades under varying conditions or long-time usage. Recent advances in deep learning-based inertial tracking have demonstrated the potential to improve performance by leveraging architectures such as Recurrent Neural Networks (RNNs) [4], [5], [6], [7] and Transformer-based models [8], [9]. Albeit with improved accuracy, they still struggle with prolonged tasks and usage in varying, complex environments. This is because neither the RNNs nor the transformers can solve the notorious drifting errors from the accelerometer and gyroscope. RNN-based models are good at capturing temporal features but suffer from operational inefficiencies and performance decreased by vanishing gradients [10]. Transformer-based models [8], [9] achieve better performance but face challenges in capturing long-term dependencies and generalizing to unseen settings.

To address these limitations, we present a novel **neural inertial tracking** (NEURIT) system that innovatively uses magnetometer data for inertial learning through a sensor fusion approach, effectively providing drift compensation.

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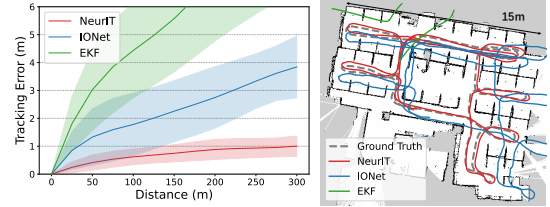


Fig. 1: Tracking performance comparison and visualization of estimated trajectories. NEURIT achieves consistent and stable tracking performance over time.

Our insight is that magnetometers, while susceptible to interference, exhibit substantially less drift than gyroscopes, thus making them a valuable resource for long-term inertial tracking. Specifically, magnetometers provide pose information, whereas accelerometers and gyroscopes offer detailed insights into motion dynamics. Based on this observation, NEURIT strategically incorporates the *differentiation of body-frame magnetometer data* to synchronize these sensor modalities, enabling enhanced orientation calibration and more robust trajectory estimation over time. Furthermore, NEURIT introduces a hybrid Time-Frequency Block-recurrent Transformer TF-BRT architecture, which uniquely combines the strengths of both RNNs and Transformers. This module captures essential features in both the temporal and frequency domains to learn complex motion dynamics while combining magnetometer data for drift compensation over extended periods.

We evaluate NEURIT using a customized robotic platform and validate its performance across two public benchmarks as well as a newly collected dataset. The results demonstrate that NEURIT significantly outperforms baseline models (see Fig. 1), achieving a 48.21% improvement in accuracy on unseen data and an average drift rate of 0.62%. Additionally, NEURIT matches or exceeds the performance of commercial visual-inertial platform (Tango Phone) [11] in both featureless and visually rich environments, rendering it a robust complementary solution for advanced tracking applications.

II. RELATED WORKS

Neural Inertial Tracking: IONet [5] first integrates neural networks for inertial tracking, which greatly improves the accuracy compared with traditional methods [12], [7]. With its success, RONIN [4] and DeepVIP [13], *etc.*, continue to explore the limit of RNN for inertial tracking and achieve promising results. After that, CTIN [8], A2DIO [9], *etc.*, replicate the success of the Transformer in the NLP field and further improve the tracking accuracy. To the best of our knowledge, NEURIT is the first work to combine the

advantages of RNN and Transformer for neural inertial tracking.

Magnetometers Usage: Most aforementioned neural inertial tracking works use accelerations and gyroscopes exclusively since magnetometers will be affected by indoor environments [14]. Some works, like A^3 [15] and MUSE [12], utilize magnetometers to calibrate gyroscope drift. In addition, Tlio [16] incorporates deep learning to enhance the robustness of EKF. However, the fundamental idea of these works under the framework of filtering techniques, which is distinct from NEURIT. NEURIT utilizes *differentiation of body-frame magnetometers* to achieve fusion and complementarity of IMU data from a neural network input perspective.

Fused Inertial Tracking: Despite advances in neural networks, accumulative path errors remain a challenge in indoor tracking. Integrating indoor maps is key to achieving high-precision localization. Approaches like NILoc [17] use transformers for location prediction from velocity sequences, while others [18] combine spatial map embeddings with IMU data. Methods incorporating IMU with RF signals [19], [20], [21] show limited improvements due to interference. Visual [22], [23] and LiDAR-based [24] solutions perform well but drift in feature-sparse areas. To improve robustness, IMU data has been integrated into visual-inertial [25] and LiDAR inertial [26], [27] systems.

III. OBSERVATIONS AND MEASUREMENTS

IMU-based tracking is often affected by orientation drift, primarily due to gyroscope noise. While outdoor environments allow global magnetometers to partially mitigate this drift, indoor settings introduce challenges, as ferromagnetic materials in buildings can distort magnetometer readings [28], [14]. This distortion likely explains the exclusion of magnetometers in prior neural inertial tracking studies. To investigate, we collect magnetometer and gyroscope data from various indoor environments and evaluate their heading orientation estimates. The heading errors are compared over time, as shown in Fig. 2. Despite the influence of building materials, global magnetometers exhibit less long-term orientation drift compared to gyroscopes. This finding suggests that magnetometers could serve as a valuable compensatory feature for neural inertial tracking.

Nevertheless, how to encapsulate the three modalities remains a challenge, given their unique perspectives on measuring the target's motion. First, for precise floor plane tracking, converting accelerations and gyroscopes from the IMU body frame to the Earth frame is crucial since the IMU operates in a non-inertial coordinate system. Unlike accelerometers and gyroscopes, which quantify motion magnitude without directly indicating orientation, magnetometers provide direct orientation information. To better align these data sources, we introduce the *derivative of magnetometers in the body frame*, which combines pose information with orientation dynamics. This improves the network's capacity to learn detailed motion patterns and serves as a critical element in our sensor fusion approach.

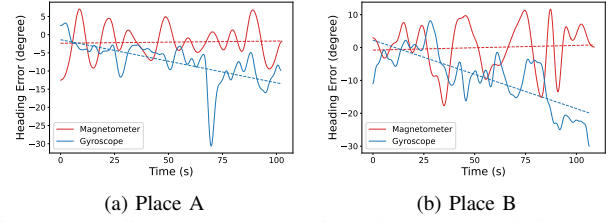


Fig. 2: Heading error comparison between global magnetometers and gyroscopes. The global magnetometer demonstrates minimal accumulation of heading errors.

IV. OUR METHOD

NEURIT employs a sequence-to-sequence framework to reconstruct robot trajectories using only IMU sensor data. Fig. 3 illustrates the system's architecture. IMU readings, including accelerations, gyroscopes, and magnetometers, are pre-processed into time series data for input. Then, TF-BRT, which is the core of NEURIT, leverages both current and historical readings to estimate the robot's current velocities, which are then integrated over time to determine its position. The design of each component is illustrated below.

A. Data Preparation

IMU sensors report raw data in the constantly changing body coordinate frame, whereas the Earth coordinate frame defines the robot's moving trajectory. To align these frames, coordinate transformation is essential. A common approach in neural inertial tracking [4] involves transforming both IMU data and ground truth into the navigation frame. In NEURIT, we adopt the same heading-agnostic transformation as RONIN [4], using the Earth coordinate frame to represent the IMU data.

The transformation is applied to accelerometer and gyroscope data: $(\mathbf{a}^g, \mathbf{w}^g) = \mathbf{q}(\mathbf{a}^b, \mathbf{w}^b)\mathbf{q}^*$, where $\mathbf{a}$ and $\mathbf{w}$ denote accelerations and gyroscopes, and b and g refer to the body frame and Earth frame, respectively. Conversely, we keep the magnetometers in the body frame and apply differentiation:

$$\begin{aligned} \dot{\mathbf{m}}^b &= \dot{\mathbf{R}} \cdot \mathbf{m}^g \\ &= \boldsymbol{\Omega} \cdot \mathbf{R} \cdot \mathbf{m}^g \\ &= \begin{pmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{pmatrix} \cdot \mathbf{R} \cdot \mathbf{m}^g, \end{aligned} \quad (1)$$

where $\dot{\mathbf{m}}^b$ is the time derivative of the magnetometer vector in the body frame, $\mathbf{R}$ is the rotation matrix from the Earth frame to the body frame, and $\boldsymbol{\Omega}$ is the angular velocity matrix formed from the gyroscope readings. This formulation enables the magnetometer's differentiation to provide real-time orientation, especially in cases where gyroscope drift accumulates over time, offering a drift-free reference for robust neural inertial tracking.

The three-modality data, $\mathbf{X} = [\mathbf{a}^g; \mathbf{w}^g; \dot{\mathbf{m}}^b]$, is then segmented using a sliding window with length L . This generates data samples $\mathbf{X}_i \in \mathbb{R}^{L \times D}$, where D equals 9 is the IMU data dimension.

B. Model Architecture

We propose TF-BRT to improve the learning outcome for IMU-based robot tracking. TF-BRT comprises six key components, as shown in Fig. 3: time-frequency input projection, block-recurrent attention module, convolution module, multi-head attention module, feedforward module, and output projection layer.

The time-frequency input projection captures essential features from both time and frequency domains, enabling a better representation of the robot's motion. The block-recurrent attention module integrates RNN into the Transformer framework, facilitating pre-sequence extraction and contextual understanding of motion within the receptive field, which improves the tracking accuracy. To expedite network convergence, the remaining four modules are connected using residual connections [29] with pre-norm residual units [30]. The convolution module employs depth-wise separable convolution [31] between layer normalization and dropout, efficiently capturing local sequence features. The multi-head attention module applies self-attention with layer normalization and dropout, while the feedforward module uses two fully connected layers to extract final features. The output projection layer distills velocity information from the high-dimensional space, producing the velocity sequence $\mathbf{V}_n \in \mathbb{R}^{L \times 2}$ on the 2D-floor plane.

This design excels in handling long sequences and demonstrates strong generalizability across various environments.

1) *Time-Frequency Input Projection*: Recent studies in time series prediction and classification highlight the value of incorporating frequency domain information, showing that patterns in time series can be more effectively captured through this domain [32], [33].

Expanding on this concept, TF-BRT employs a dual-channel input system that integrates features from both the time and frequency domains. The Discrete Cosine Transform (DCT) is used to extract frequency domain information, thereby circumventing the necessity for complex number computations within neural networks.

The two channels are connected by projection layers, which extract hidden features through the fully connected layer and map the IMU feature domain ($\mathbb{R}^{L \times D}$) to a larger domain ($\mathbb{R}^{L \times D_h}$), enabling subsequent networks to learn and operate within the expanded high-dimensional feature space. D_h represents the feature dimension in the following modules. The data after projection are merged to combine the extracted high-dimension features of the time domain and frequency domain. This operation is formalized as $f_{\theta_t}(\mathbf{X}_n) + f_{\theta_f}(\text{DCT}(\mathbf{X}_n)) = \mathbf{M}_n$, where θ_t and θ_f are the projection layer parameters for the time and frequency channels, and $\mathbf{M}_n$ represents the merged hidden features.

2) *Block-Recurrent Attention Module*: Our model builds on the block-recurrent transformer framework [34], combining the benefits of both RNNs and self-attention mechanisms. To further enhance the temporal order representation, we incorporate relative position encoding [35], which improves the model's ability to capture sequential dependencies. The block-recurrent attention module follows a pre-norm

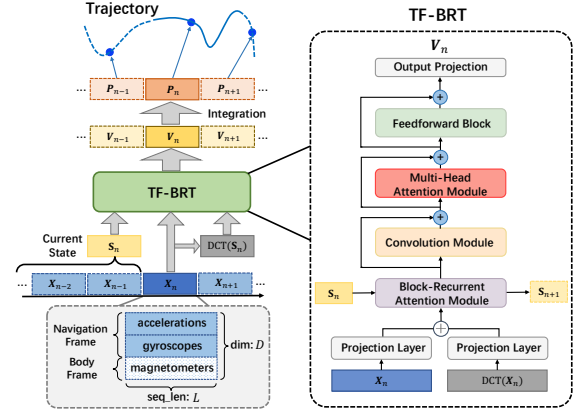


Fig. 3: NEURIT Overview. Left side: End-to-end indoor tracking framework; Right side: TF-BRT architecture.

residual structure. Specifically, the merged hidden features $\mathbf{M}_n$ are first normalized using layer normalization: $\mathbf{E}_n = \text{LayerNorm}(\mathbf{M}_n)$.

The block-recurrent attention layer then receives two inputs: the normalized embedding $\mathbf{E}_n \in \mathbb{R}^{L \times D_h}$, which represents the current IMU data, and the current state $\mathbf{S}_n \in \mathbb{R}^{L \times D_h}$, which encapsulates information from previous sequences. To generate the next output embedding $\mathbf{E}_{n+1}$ and state $\mathbf{S}_{n+1}$ (illustrated in Fig. 4a and Fig. 4b), we first compute the concatenation of cross-attention between $\mathbf{E}_n$ and $\mathbf{S}_n$ and multi-head self-attention [36] on $\mathbf{E}_n$:

$$\mathbf{H}_n = \text{Cat}(\text{CrossAttn}(\text{RPE}(\mathbf{E}_n), \text{RPE}(\mathbf{S}_n)), \text{MultiAttn}(\text{RPE}(\mathbf{E}_n))) \quad (2)$$

where RPE denotes relative position encoding. After a projection layer, $\mathbf{H}_n$ is merged with $\mathbf{E}_n$, and the final output embedding $\mathbf{E}_{n+1}$ is produced using a MLP in a residual format: $\mathbf{E}_{n+1} = \text{Res}(\text{MLP}(\text{Proj}(\mathbf{H}_n) + \mathbf{E}_n))$.

Fig. 4b outlines the workflow for generating the next state, $\mathbf{S}_{n+1}$. The primary distinction is the use of an LSTM gate, which enables the selective integration of present and historical data, essential for predicting future states. All tensors operate within the space $\mathbb{R}^{L \times D_h}$. Finally, $\mathbf{E}_{n+1}$ is passed to the next module after applying a dropout layer.

C. Multi-Loss Learning

To enhance TF-BRT's regression performance in estimating velocities and heading orientations, we implement a multi-loss learning strategy. This approach integrates three components: velocity loss $\mathcal{L}_v$, position loss $\mathcal{L}_p$, and orientation loss $\mathcal{L}_o$.

Velocity Loss ($\mathcal{L}_v$): With a predicted velocity sequence $\mathbf{V}_n = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_L]$, a mean square loss is utilized to measure the difference between the ground truth velocity $\mathbf{V}_n^g = [\mathbf{v}_1^g, \mathbf{v}_2^g, \dots, \mathbf{v}_L^g]$. The velocity loss function can be written as $\mathcal{L}_v = \sqrt{\frac{1}{L} \sum_{i=1}^L (\mathbf{v}_i - \mathbf{v}_i^g)^2}$.

Position Loss ($\mathcal{L}_p$): The predicted velocity is integrated to obtain the moving trajectory $\mathbf{P}_n$. By using $\mathcal{L}_p$, we aim to further reduce the cumulative error based on the learned

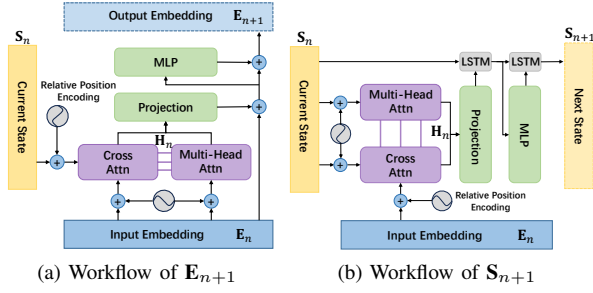


Fig. 4: Block-Recurrent Attention Layer

velocity characteristics. The equation for $\mathcal{L}_p$ is $\mathcal{L}_p = \sqrt{\frac{1}{L} \sum_{i=1}^L (\mathbf{p}_i - \mathbf{p}_i^g)^2}$.

Orientation Loss ($\mathcal{L}_o$): Orientation deviations, particularly at low speeds, can result in significant errors even with small velocity discrepancies. To address this, we introduce orientation loss to improve the accuracy of velocity direction prediction: $\mathcal{L}_o = \sqrt{\frac{1}{L} \sum_{i=1}^L (\frac{\mathbf{v}_i}{\|\mathbf{v}_i\|_2} - \frac{\mathbf{v}_i^g}{\|\mathbf{v}_i^g\|_2})^2}$.

The final multi-loss function combines these losses using the variation coefficient [37] to balance the convergence process:

$$\mathcal{L}_{total} = \frac{\sigma_{\mathcal{L}_v}}{\mu_{\mathcal{L}_v}} \mathcal{L}_v + \frac{\sigma_{\mathcal{L}_p}}{\mu_{\mathcal{L}_p}} \mathcal{L}_p + \frac{\sigma_{\mathcal{L}_o}}{\mu_{\mathcal{L}_o}} \mathcal{L}_o \quad (3)$$

where μ and σ denote the mean and standard deviation of the respective loss function. By combining velocity loss $\mathcal{L}_v$ and position loss $\mathcal{L}_p$, the model applies corrections at both the frame level and over entire time windows, improving its ability to capture the motion dynamics.

D. Data Augmentation

To improve the robustness of TF-BRT under different heading directions, both IMU data $\mathbf{X}_n$ and ground truth velocity $\mathbf{V}_n^g$ are randomly rotated by an angle of $\phi \in [0, 2\pi)$ on the 2D-floor plane [38]: $\Delta \mathbf{q} \mathbf{V}_n^g \Delta \mathbf{q}^* = \Delta \mathbf{q} \mathbf{X}_n \Delta \mathbf{q}^*$, where $\Delta \mathbf{q}$ represents the random rotation in quaternion. By taking this data augmentation method, the moving orientation will distribute more evenly on the floor plane, thus making the model more robust.

V. IMPLEMENTATION

We build a customized robot platform for data collection, which captures raw IMU data as NEURIT input, visual-inertial tracking results from a commercialized device (Tango Phone) for comparison, and the ground truth data from a LiDAR-inertial navigation system based on FAST-LIO [26]. The Tango phone is selected due to its established use as a ground truth reference in inertial tracking datasets such as RIDI and RONIN. FAST-LIO is chosen for its minimal drift in indoor environments, making it a reliable source for accurate ground truth measurements.

Fig. 5a illustrates the system setup. The main component of the system is an Intel NUC with an i7-1260P CPU. By obtaining point clouds through LIVOX MID360 and the built-in IMU, the FAST-LIO algorithm, which runs on the NUC, can accurately determine the indoor location of the

TABLE I: Statistical Summary of Datasets

Dataset	Year	IMU Carrier	Sample Frequency	# of Sequences	Ground Truth
NeurIT	2023	CH110	200 Hz	110	FAST-LIO
RIDI	2017	Lenovo Phab2 Pro	200 Hz	98	Google Tango phone
RONIN	2019	Galaxy S9, Pixel 2 XL	200 Hz	118	Asus Zenfone AR

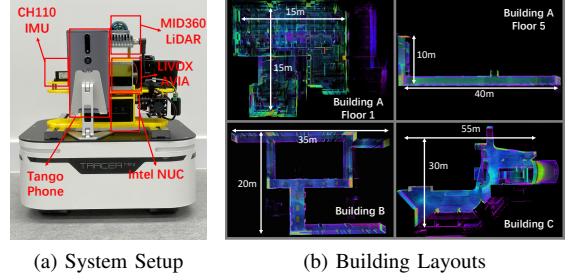


Fig. 5: Experimental Settings. The floorplans are visualized based on the results reconstructed by FAST-LIO.

robot. HiPNUC CH110 9-axis IMU is set on the left side, which records the raw data for model input. The Tango Phone set in the front is used to collect visual-inertial tracking results for comparison in §VII-D. In addition, we add AVIA (FOV: $70.4^\circ \times 77.2^\circ$), a narrow-FOV LiDAR, to demonstrate the fact that LiDAR-inertial tracking systems cannot work in simple and plain indoor environments. The robot's territory is AgileX Tracer-Mini, whose size is $40\text{cm} \times 40\text{cm} \times 26\text{cm}$ with a maximum speed of 1.5m/s.

VI. EXPERIMENTS

A. Datasets

Due to the lack of IMU datasets on robots subject, we build our dataset to evaluate TF-BRT's effectiveness. The dataset is split into four parts: training, validation, test-seen, and test-unseen sets. Details are as follows:

NEURIT: The NEURIT dataset is collected using a customized robotic platform across four floors in three buildings, each with different layouts, environments, and floor conditions (see Fig. 5b). Data for the training, validation, and test-seen sets are collected in Building A, while test-seen and test-unseen sets are built from Buildings B and C. The robot moves at varying speeds, with up to 1.5 m/s. The dataset includes 110 sequences, totaling approximately 15 hours of tracking data covering a distance of 33.7 km. Each sequence lasts between 6 and 10 minutes. The split ratio for the training, validation, test-seen, and test-unseen sets is 15:3:3:4.

To further evaluate the robustness of NEURIT, we assess its performance on two widely used pedestrian inertial tracking datasets: RIDI and RONIN. Since only a subset of the RONIN dataset has been publicly released, our reported results may differ from those in the original paper. The statistical summary of the datasets is detailed in Tab. I.

During training, we segment NEURIT and RIDI using a window size of $200\text{hz} \times 15\text{s}$ with a step size of 400. For RONIN, we use a window size of $400\text{hz} \times 20\text{s}$ with a step

TABLE II: Evaluation on NEURIT Dataset. NEURIT surpasses the best baseline by an average of 29.3%.: (The best results among all algorithms are highlighted in **bold**, and the best baseline is shown in underline format.)

Dataset	Subject	Metric	NDI	EKF	RONIN	IONet	UniTS	RF-Net	NeurIT w/o magn	Impvt	NeurIT	Impvt
NeurIT Dataset	Seen	ATE	7.93	7.65	3.59	3.26	<u>2.71</u>	5.35	1.53	43.56% ↑	1.01	62.89% ↑
		RTE	10.01	9.61	2.19	1.58	<u>1.37</u>	2.09	1.01	26.18% ↑	0.58	57.31% ↑
		AYE	123.09	124.77	63.77	<u>46.23</u>	121.41	99.76	37.90	18.02% ↑	35.99	22.15% ↑
	Unseen	ATE	9.24	8.96	4.26	3.45	<u>2.90</u>	4.12	1.77	39.11% ↑	1.14	60.89% ↑
		RTE	11.24	11.10	2.42	1.87	<u>1.74</u>	2.24	1.61	7.09% ↑	0.76	56.42% ↑
		AYE	132.90	134.11	51.66	<u>40.10</u>	119.55	89.36	33.58	16.25% ↑	32.14	19.86% ↑
RIDI	Seen	ATE	16.75	26.04	1.72	1.76	<u>1.66</u>	3.94	1.63	1.40% ↑	1.53	7.64% ↑
		RTE	17.58	37.51	<u>2.04</u>	2.09	<u>2.16</u>	4.34	2.04	0.16% ↑	2.02	1.19% ↑
		AYE	127.97	114.15	<u>51.39</u>	57.52	70.95	63.74	27.21	47.05% ↑	26.33	48.76% ↑
	Unseen	ATE	17.30	24.14	<u>1.87</u>	2.10	1.99	3.31	1.51	19.10% ↑	1.51	19.36% ↑
		RTE	18.02	40.67	<u>1.87</u>	2.18	1.92	3.20	1.66	11.16% ↑	1.60	14.50% ↑
		AYE	130.59	105.93	<u>48.47</u>	53.67	67.42	64.64	24.34	49.78% ↑	24.21	50.05% ↑
RONIN	Seen	ATE	22.86	33.26	7.76	6.20	<u>5.68</u>	8.78	4.58	19.38% ↑	4.37	23.10% ↑
		RTE	30.49	35.10	3.19	3.17	<u>2.77</u>	6.43	2.60	6.03% ↑	2.59	6.53% ↑
		AYE	122.48	120.96	<u>63.12</u>	74.49	107.92	86.91	46.56	26.24% ↑	47.60	24.59% ↑
	Unseen	ATE	17.30	33.47	8.91	7.73	<u>6.38</u>	10.27	5.78	9.52% ↑	5.35	16.26% ↑
		RTE	18.02	41.07	4.12	4.63	<u>3.86</u>	8.20	3.59	7.12% ↑	3.55	8.05% ↑
		AYE	110.00	114.80	<u>64.46</u>	76.49	106.80	88.69	47.88	25.72% ↑	46.66	27.61% ↑

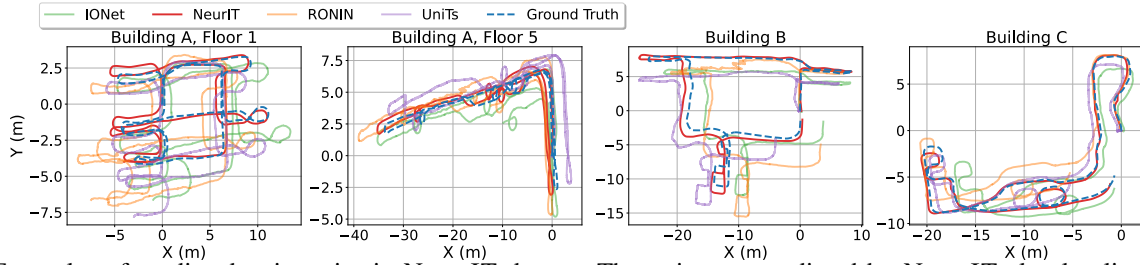


Fig. 6: Examples of predicted trajectories in NEURIT dataset. The trajectory predicted by NEURIT closely aligns with the ground truth, whereas baseline methods exhibit increasing deviation over time.

size of 1000. During testing, we apply the same window size and step size of 200 for NEURIT dataset and RIDI and 400 for RONIN.

B. Baselines

To comprehensively evaluate NEURIT, we compare its performance against six state-of-the-art baselines. These include two traditional control algorithms: Naive Double Integration (NDI [39]) and Extended Kalman Filter (EKF [2]); two neural inertial tracking algorithms: IONet [5] and RONIN [4]; and two time-frequency learning models: UniTS [32] and RF-Net [33].

Transformer-based models, such as CTIN [8] and A2DIO [9], have shown promising results in inertial tracking. However, due to the lack of implementation details and publicly available source code, we cannot reproduce these results. Therefore, they are not included as baselines. Performance comparisons with Transformer-based methods will be shown in the ablation study (§VII-B).

C. Evaluation Metrics

We consider three common criteria to assess the error between the estimated positions $\mathbf{P}$ and the ground truths $\mathbf{P}^g$.

- **Absolute Trajectory Error (ATE):** ATE [40] calculates the root mean square error (RMSE) of the predicted trajectory and the ground truth trajectory.

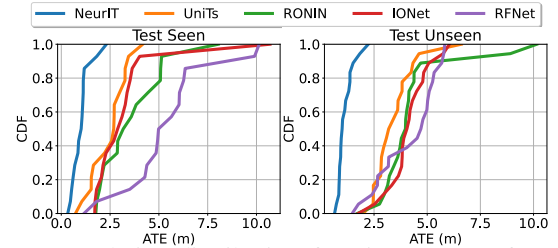


Fig. 7: Cumulative distribution function (CDF) of ATE on NEURIT test set. NEURIT notably outperforms baselines, particularly in unseen environments.

- **Relative Trajectory Error (RTE):** RTE [40] measures the RMSE between the predicted trajectory and the ground truth over a time-interval Δt (60s in our study).
- **Absolute Yaw Error (AYE):** AYE [41] measures the RMSE between the heading orientations between ground truth and predicted trajectories.

VII. EVALUATION

A. Overall Performance

Tab. II demonstrates NEURIT's strong performance on NEURIT dataset. NEURIT reduces ATE by 62% compared to the best baseline and achieves an RTE of 0.6%, nearly halving the baseline result. With multi-loss learning, the

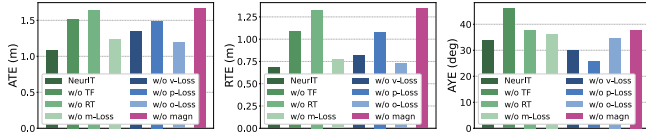


Fig. 8: Ablation Study. Green bars: TF-BRT modules; Blue bars: Loss functions; Purple bar: Magnetometers.

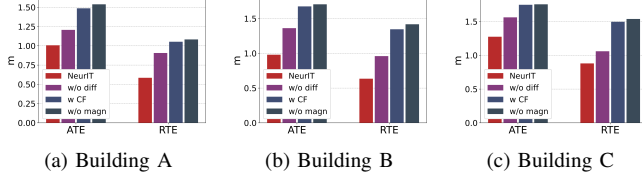


Fig. 9: Magnetometer study. **diff**: Differential operation; **CF**: Implementation of a complementary filter.

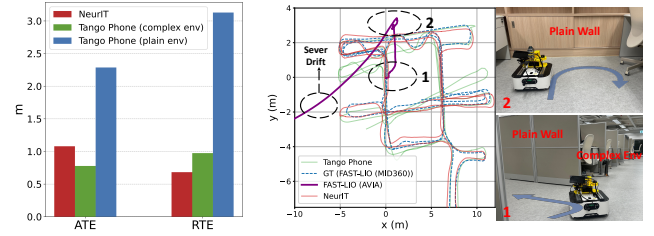
heading error drops by 20%, from 43° to 33° . Even without magnetometers, NEURIT outperforms baselines by 30% across three metrics, and with magnetometers, RTE improves by 50%. The results also underscore NEURIT's effectiveness in pedestrian tracking, with a 17% improvement on RIDI's seen dataset and a 25% gain on the unseen dataset. While the performance gains on the RONIN dataset are modest, possibly because the RONIN dataset only released half of the total materials, NEURIT consistently outperforms all baselines, highlighting its potential as a unified model for both robotic and pedestrian tracking. To provide a more comprehensive evaluation, we include the cumulative distribution function (CDF) of ATE in Fig. 7. Notably, NEURIT showcases a more stable capacity with much lower variance. The performance gains are even more pronounced in unseen environments. Additionally, Fig. 6 illustrates the predicted trajectories, showing that NEURIT aligns closely with the ground truth over time and distance. In contrast, other methods exhibit severe deviations, further emphasizing NEURIT's superior accuracy and reliability.

B. Ablation Study

We evaluate the effectiveness of different modules in NEURIT, focusing on time-frequency input (TF), block-recurrent attention (BR), and multi-loss learning (m-loss), as seen in Fig. 8. Replacing the TF with time-domain channels and BR with self-attention degrades performance, highlighting the importance of these components. Multi-loss learning proves effective, as the single-loss function reduces accuracy. The best results come from combining the three loss functions, balancing velocity, trajectory, and orientation errors.

C. Magnetometer Study

In NEURIT, we propose using *differentiation of body-frame magnetometers* for orientation compensation. To validate this component, we compare three approaches: using magnetometers without differentiation, applying a magnetometer-based complementary filter (CF) for drift compensation [12], [7], and using our method. Tests on 200m+ trajectories across three buildings show our method



(a) Comparison on ATE, RTE

(b) Trajectory comparison

Fig. 10: Comparison with Tango Phone and narrow-FOV LiDAR (AVIA). We compare NEURIT with Tango Phone across various visual settings as shown in (a), and present instances where the narrow-FOV LiDAR-inertial tracking system (AVIA) fails in extremely plain environments in (b).

achieves the best results (Fig. 9). The undifferentiated approach sees a 30% drop, while the CF approach offers only a 4% improvement over not using magnetometers, rendering its limitations in complex scenarios.

D. Comparison with Tango Phone and AVIA

We further compare NEURIT with the Tango phone's visual-inertial algorithm and the FAST-LIO algorithm using a narrow FOV LiDAR (AVIA). In feature-rich environments, NEURIT performs similarly to the Tango phone, with an ATE of 1.08m versus 0.78m. However, in plain areas, NEURIT considerably outperforms the Tango phone. We do not report AVIA-based FAST-LIO results in visually constrained environments due to significant drift of hundreds of meters. Fig. 10 illustrated the predicted trajectories of each method, the Tango phone drifts at corners with no visual points, while NEURIT stays close to the ground truth. AVIA-based FAST-LIO experiences severe drift, especially after plain walls, making accurate trajectory tracking impossible.

VIII. CONCLUSION

In this paper, we present NEURIT, a neural inertial tracking system that fully leverages IMU data. We introduce a novel Time-Frequency Block-recurrent Transformer, TF-BRT, which combines time-frequency learning with the strengths of both RNNs and Transformers to integrate historical and current motion information. Additionally, we employ body-frame magnetometer differentiation for drift compensation. We implement NEURIT with a customized robotic platform and evaluate it across distinct indoor environments. The results prove its robustness, even in previously unseen settings. To foster further research in this field, we open-sourced the collected dataset and code implementations. We believe this research represents a significant step toward practical, infrastructure-free indoor robotic tracking.

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